

# Software Design and Architecture

[Architecture Patterns] – Chapter 04, L03

---

# Lecture Outlines

## ➤ Interactive Systems

- ✓ Overview

- ✓ Patterns:

  - Model-View-Controller (MVC).

## ➤ Hierarchical Systems

- ✓ Overview

- ✓ Patterns:

  - Layered

# Interactive Systems

- Interactive systems support user interactions, typically through user interfaces.
  - ✓ When designing these systems, two main quality attributes are of interest:
    - Usability
    - Modifiability
- The mainstream architectural pattern employed in most interactive systems is the Model-View-Controller (MVC).

# Interactive Systems

- The MVC pattern is used in interactive applications that require flexible incorporation of human-computer interfaces. With the MVC, systems are decomposed into three main types of components:

| Component  | Description                                                                                                  |
|------------|--------------------------------------------------------------------------------------------------------------|
| Model      | Component that represents the system's core, including its major processing capabilities and data.           |
| View       | Component that represents the output representation of the system (e.g., graphical output or console-based). |
| Controller | Component (associated with a view) that handles user inputs.                                                 |

Extra:

### Mobile App – Internal Architecture (MVC)

- **Model** → Manages the app's data (local storage, cached data, or data already fetched).
- **View** → User interface (screens, buttons, forms).
- **Controller** → Handles user input, communicates with services/APIs if needed, updates the Model and View.

**Note:** The **Controller (or service layer)** is usually responsible for calling the server, not the Model directly.

---

### Mobile App – External Architecture (Distributed System)

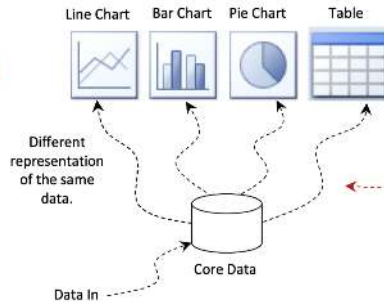
- **Client (Mobile App)** → Sends requests (login, get messages, check balance) through APIs.
- **Server** → Processes requests, applies business rules.
- **Database** → Stores persistent data (users, transactions, messages).

### Summary:

- Inside the mobile app = **MVC** organizes the code (UI, logic, data).
- Outside the app = **Distributed System** (Client ↔ Server ↔ Database) enables communication and services.

# MODEL-VIEW-CONTROLLER ARCHITECTURAL PATTERN

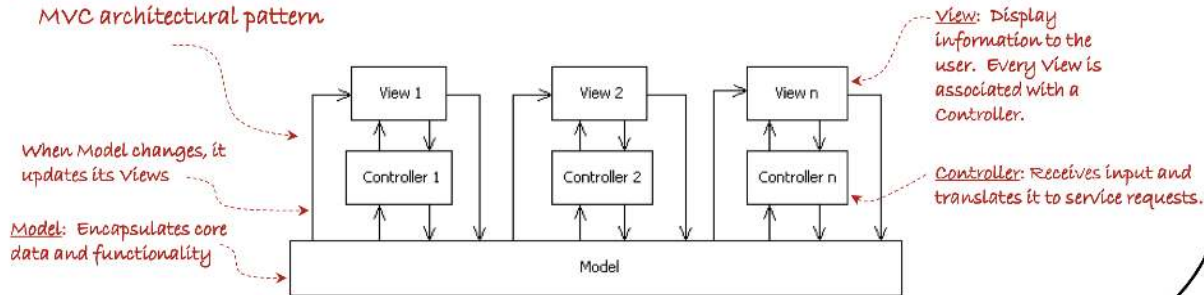
*Consider the popular example where data needs to be represented in different formats*



*When data changes, all views are updated to reflect the changes.*

# MODEL-VIEW-CONTROLLER ARCHITECTURAL PATTERN

Box-and-line diagram of the MVC architectural pattern



## Extra:

### 1. Model

1. Holds core data and system functionality.
2. When data changes, it updates the Views.

### 2. View

1. Displays information to the user.
2. Each View is linked with a Controller.

### 3. Controller

1. Receives user input.
2. Translates input into service requests for the Model.

### 4. Flow of interaction

1. User interacts with **Controller**.
2. Controller updates the **Model**.
3. Model notifies and updates the **View**.
4. View presents the updated information back to the user.

## Example Scenario: User Updates Their Profile Name

### 1. User action (input)

1. The user types a new name and clicks **Save**.
2. This action goes to the **Controller**.

### 2. Controller

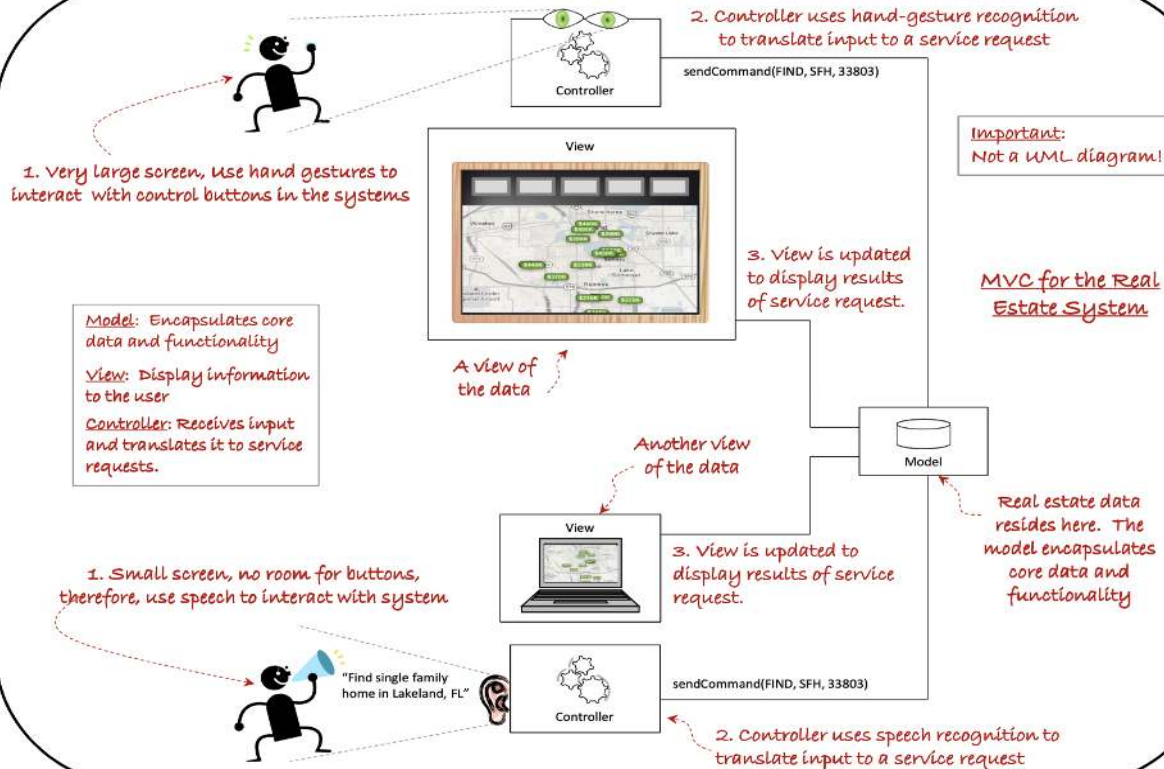
1. The Controller receives the input (new name).
2. It translates the input into a **request to update the Model**.

### 3. Model

1. The Model updates the user's name in its data.
2. After updating, the Model **notifies the View** that data has changed.

### 4. View

1. The View retrieves the updated data from the Model.
2. It refreshes the screen to display the new name.





# Extra: Explanation of the System:

## 1. User

**Large Screen:** Uses hand gestures to navigate or control options.

**Small Screen:** Uses voice commands due to limited space.

## 2. Controller

**Role:** Connects View and Model.

**Task:** Converts user input (gestures/voice) into actions like fetching or filtering property data.

## 3. Model

**Role:** Stores real estate data (properties, prices, locations).

**Task:** Provides data to the View and updates based on Controller requests.

## 4. View

**Role:** Shows information to the user.

**Task:** Updates display dynamically based on user actions (gestures/voice).

👉 In short: **User** → **Controller** → **Model** → **View** → **User** (with gestures/voice as inputs).

# MODEL-VIEW-CONTROLLER ARCHITECTURAL PATTERN

## Extra:

### 1. RealEstateModel (Model)

Central data component storing real estate information.  
Exposes **IObservable** for updates and **IModel** for data access.

### 2. Controllers

**SbController** (SmartBoard Controller): Handles pen/gesture input via **IPenController**.

**SpeechController**: Handles voice commands via **ISpeechController**.

Both translate user input into actions on the **RealEstateModel** (through **IModel**).

### 3. Views

**SmartBoardView**: Displays info for large-screen (gesture/pen-based).

**PCView**: Displays info for smaller screen (voice-based).

Both register as **IObservers** to the **RealEstateModel**, so they update automatically when the Model changes.

Note: View requires the Model (to get the data).

But the Model also requires View (to send updates).

So, both **provide** and **require** exist at the same time because of the **two-way dependency** created by the **Observer pattern**.

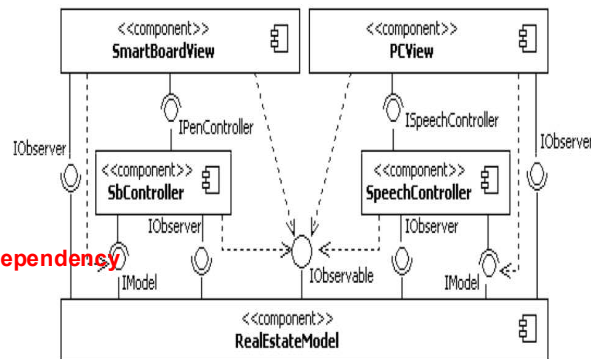
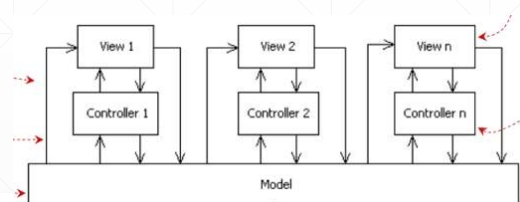
### 4. Communication Flow

User interacts → Controller (pen or speech).

Controller updates the **RealEstateModel**.

Model notifies Views via **IObserver**.

Views refresh to display updated real estate data.



👉 In short: **User input** → **Controller** → **RealEstateModel** → **Views** (auto-updated via **Observer pattern**).

### SpeechController Class Methods:

- processSpeech() (called when voice input is received)
- sendCommand(FIND\_CMD, 5FH, 33803) (sends parsed command to model)

### RealEstateModel Class Methods:

- processCommand() (processes the FIND command)
- update() (updates internal state)
- getResults() (retrieves search results)

### PCView Class Methods:

- show() (called twice - displays the view)
- *Internally enables voice input mode (not a method call, but internal state change)*

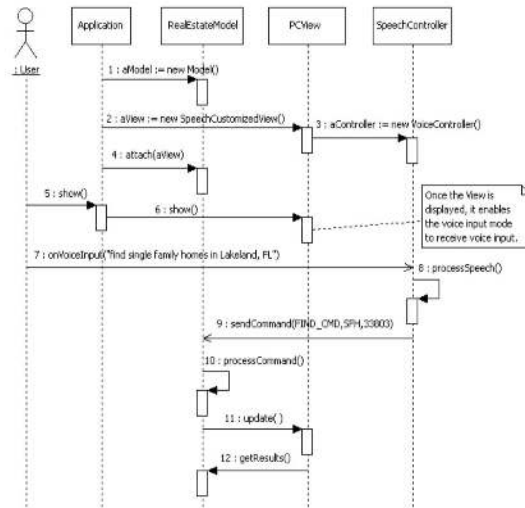
### Application Class Methods:

- new Model() (constructor)
- new SpeechCustomizedViewModel() (constructor)
- new VoiceController() (constructor)
- attach(aView) (attaches view)

### External/User:

- onVoiceInput("find single family homes in Lakeland, FL") (this is the user/system calling the controller's input handler)

## MODEL-VIEW-CONTROLLER ARCHITECTURAL PATTERN



# MODEL-VIEW-CONTROLLER ARCHITECTURAL PATTERN

- Quality properties of the MVC architectural pattern include the ones specified below.

| Quality       | Description                                                                                                                                                                        |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Modifiability | Easy to exchange, enhance, or add additional user interfaces.                                                                                                                      |
| Usability     | By allowing easy exchangeability of user interfaces, systems can be configured with different user interfaces to meet different usability needs of particular groups of customers. |
| Reusability   | By separating the concerns of the model, view, and controller components, they can all be reused in other systems.                                                                 |

- There are many variations of the MVC architectural pattern.

- ✓ One popular variation includes the fusion of views and controller components, as made famous in the 1990s by Microsoft's Document-View architecture
- ✓ Other more extensive variations include the process-abstraction-controller pattern.

## Extra:

**Modifiability** - Easy to exchange or enhance user interfaces

**Usability** - Configure different UIs for different customer needs

**Reusability** - Model, View, and Controller components can be reused in other systems

# Hierarchical Systems

- Hierarchical systems can be decomposed and structured in hierarchical fashion. Two common architectural patterns for hierarchical systems are:
  - ✓ Main program and subroutine
  - ✓ Layered

# Extra: Hierarchal Systems

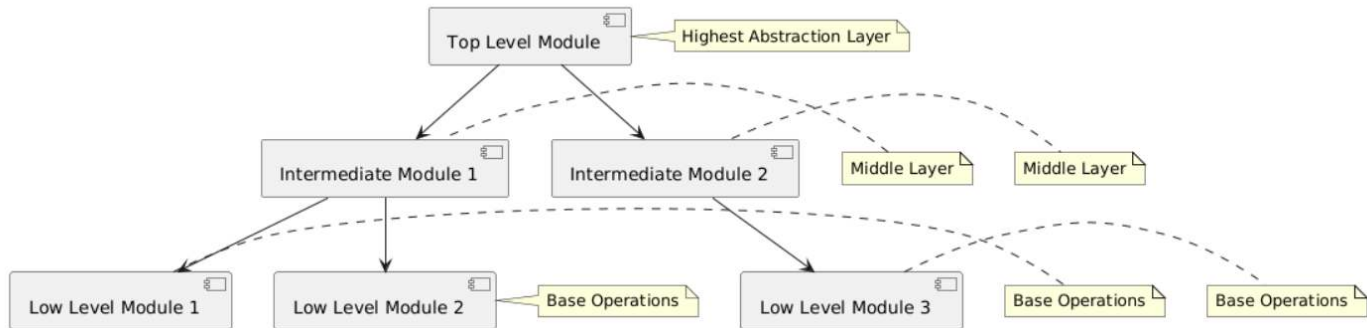
- Description:**

These systems are structured in layers, each layer having a specific role or responsibility, allowing for the separation of concerns across a vertical (layered) or horizontal (module) scale.

- Example:**

- Operating systems where different layers handle tasks ranging from hardware interaction at the lowest level to user interface management at the highest level.

## Hierarchical System



# Layered Pattern

- Quality properties of the Layered architectural pattern include the ones specified below.

| Quality       | Description                                                                                                                                                                                                                                                                                      |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Modifiability | Dependencies are kept local within layer components. Since components can only access other components through a well-defined and unified interface, the system can be modified easily by swapping layer components with other enhanced or new layer components.                                 |
| Portability   | Services that deal directly with platform's API's can be encapsulated using a system layer component. Higher level layers rely on this component for providing system services to the application, therefore, by porting the system's API layer to other platforms systems become more portable. |
| Security      | The controlled hierarchical structure of layered systems allow for easy incorporation of security components to encrypt/decrypt incoming/outgoing data.                                                                                                                                          |
| Reusability   | By compartmentalizing each layer's services, they become easier to reuse.                                                                                                                                                                                                                        |

## Extra:

**Modifiability** - Easy to swap layer components through defined interfaces

**Portability** - Encapsulate platform APIs for easier cross-platform migration

**Security** - Easy to add security components in controlled hierarchical structure

**Reusability** - Compartmentalized layer services are easier to reuse

# Layered Pattern

## SatcomLayer Component

- Purpose:** Handles satellite communication functions
- Interface:** ISatcom (**provides** satellite communication services to other layers)

## SecurityLayer Component

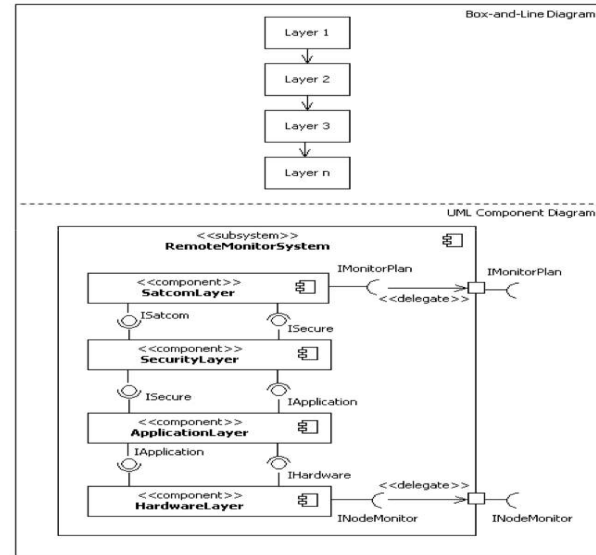
- Purpose:** Manages security and encryption functions
- Interface:** ISecure (**provides** security services, **requires** application services)

## ApplicationLayer Component

- Purpose:** Contains main application logic and business rules
- Interface:** IApplication (**provides** application services, **requires** hardware access)

## HardwareLayer Component

- Purpose:** Controls physical hardware devices and sensors
- Interface:** IHardware (**provides** hardware access, **requires** monitoring services)





## WHAT'S NEXT...

- • In this session, we continued the discussion on distributed systems and presented fundamentals concepts of interactive and hierarchical systems, together with essential architectural patterns for these systems, including:
  - ✓ MVC
  - ✓ Layered
- This finalizes our coverage of architectural patterns. In the next module, we start the discussion on detailed design, which is the next activity in the design process. Detailed design begins once the architecture of the software is sufficiently complete.